

VLAN (Virtual Local Area Network):

- o In Cisco World the VLAN stands for Virtual Local Area Network.
- o Virtual Local Area Network is a logical grouping of networking devices.
- o Break large broadcast domain in smaller broadcast domains to use VLANs.
- o Virtual Local Area Network divide the switch into different LAN segments.
- o Think like or consider Virtual Local Area Network (VLAN) as a subnet.
- o Two different subnets cannot communicate with each other without Router.
- o Two different Virtual Local Area Network also requires router to communicate.
- o Virtual Local Area Network used in Cisco Switches and it operates at OSI Layer 2.
- o By default, in every Cisco switch there is Five Virtual Local Area Network (VLANs).
- o Virtual Local Area Network (VLAN) 1 is the default VLAN for every Cisco switch port.
- o The standard Virtual Local Area Network (VLAN) range is from VLAN 1 to 1005.
- o The Extended Virtual Local Area Network (VLAN) range is from VLAN 1006 – 4094.
- o The Extended-range are store in running configuration of every Cisco Switches.
- o In Cisco VTP version 3 and transparent mode support extended-range in Switches.
- o The reserved Virtual Local Area Network (VLAN) range is from VLAN 1002-1005.
- o VLANs 1, 1002 to 1005 are created automatically and cannot be removed or assign.
- o Voice Virtual Local Area Network enables the access port to carry IP voice traffic.
- o By default, in every cisco Switch the voice Virtual Local Area Network is disabled.
- o Virtual Local Area Network are stored in Cisco Switch **vlan.dat** file in Flash Memory.

VLAN Analogy:

- o Partition hard disks on a computer into several logical partitions.
- o The main purpose of hard disks partition is to organize the disk space.
- o It ensure that corruption of partition does not affect another partition.
- o Like a big room separated by partitions to make them two houses for use.
- o VLAN is like as big office separated by partitions makes in many offices.
- o It is like a big classroom, divide into many section such as Class A, B, C etc.

Advantage of VLAN:

- o Solve broadcast problem.
- o Reduce the size of broadcast domains.
- o Allow us to add additional layer of security.
- o Make device management easier.
- o Implement logical grouping of devices by function.

Type of VLAN	Descriptions
Data	Use for normal data
Voice	Use for IP phone/Voice over IP
Private	Use for security, it divided primary and secondary VLAN
Management	A VLAN which is used for Telnet and SSH configuration
Extended	Usable if VTP mode is transparent on switches
Native VLAN	Untagged frame is send over trunk link (by default VLAN1)

VLAN Database:

- o The VLAN database is used to store vlan data, such as the VLAN ID, name and MTU.
- o The default location of the Virtual Local Area Network database is in local **vlan.dat** file.
- o Virtual Local Area Network database is stored in non-volatile memory, which is flash.

Commands	Description
SW1#dir SW1#show flash	Verify VLAN database
SW1#more unix:/vlan.dat-00004	To read VLAN database
SW1#delete unix:/vlan.dat-00004	Delete VLAN database

```
SW1#dir
Directory of unix:/

5111993  -rw-      1136   Feb 2 2019 10:03:09 +00:00  NETMAP
5111994  -rw-     16384   Feb 2 2019 10:03:09 +00:00  nvram_00004
5111900  -rw-      1373   Feb 2 2019 09:04:34 +00:00  startup-config.cfg
5112796  -rw-       616   Feb 2 2019 11:00:25 +00:00  vlan.dat-00004
5112791  -rw-       787   Feb 2 2019 10:03:09 +00:00  ubridge.log
```

Normal VLAN:

- o Normal Virtual Local Area Network range are from VLANs **1** to VLAN **1005**.
- o Normal VLANs can be configured both database configuration mode & global.
- o Normal Virtual Local Area Network are stored in **vlan.dat** file in Flash memory.
- o VTP versions 1 and 2 can advertise normal range Virtual Local Area Network only.

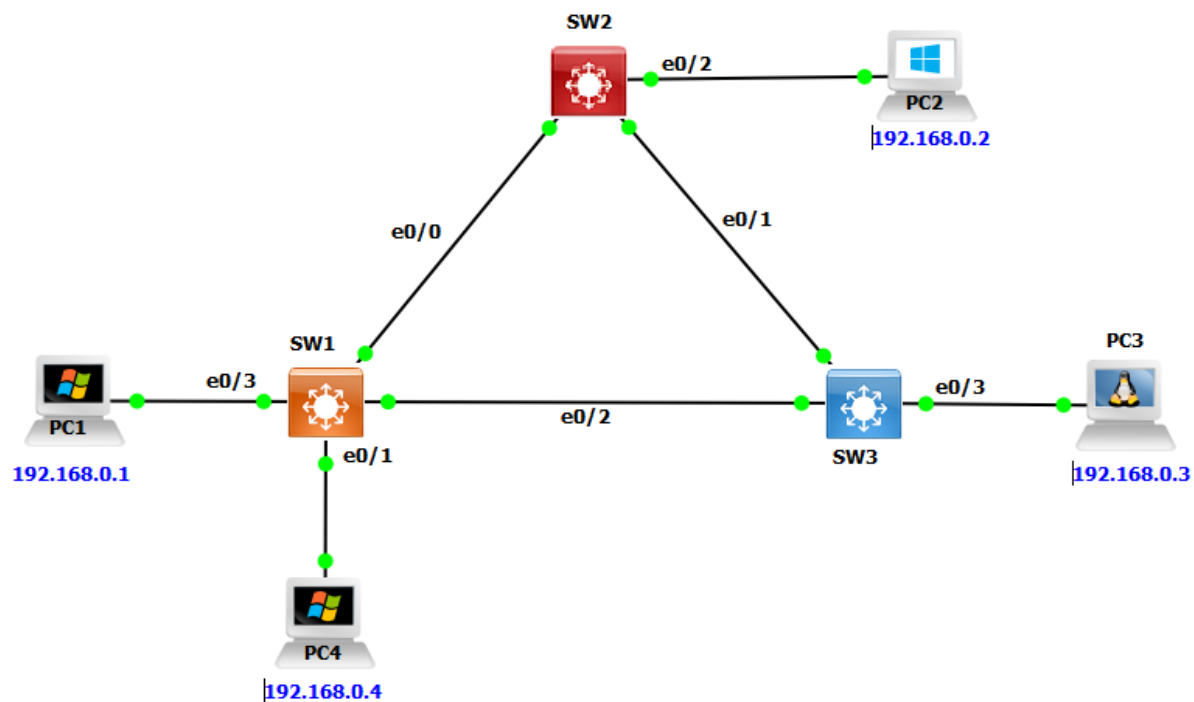
Extended VLANs:

- o Extended Virtual Local Area Network are VLANs that fall in the range **1006 to 4094**.
- o Extended Virtual Local Area Network are mainly used in service provider networks.
- o Extended VLAN allow for the provisioning of large numbers of customers in ISP.
- o Extended VLANs differ from normal VLANs because they have higher numbers.
- o Extended VLANs must be configured in VTP transparent mode or VTP Version 3.
- o Extended VLANs are saved and store in Cisco Switch running-configuration.

```
SW1(config)#vlan 1008
SW1(config-vlan)#exit
SW1(config)#exit
SW1#show running-config | sec vlan
vlan internal allocation policy ascending
vlan 1008
```

Voice VLAN:

- o Voice VLAN enables access port to carry IP voice traffic from an IP phone.
- o Voice VLAN feature enables access ports to carry IP voice traffic.
- o The voice VLAN feature is supported only on access ports.
- o The Voice VLAN is also known as the Auxiliary VLAN (AUX VLAN).
- o By default, in Cisco Switch the voice VLAN is disabled.



Commands	Description
SW1(config)#vlan 10	Create a VLAN
SW1(config-vlan)#name HR	Name VLAN
SW1(config)# no vlan 10	Delete VLAN
SW1# show vlan brief	Verify VLAN creation
SW1# show flash OR dir	Verify VLAN database
SW1(config)#vlan 5	Configure a Voice VLAN
SW1(config-vlan) # name Voice	Name voice VLAN
SW1(config)#interface e0/1	Interface mode
SW1(config-if)#switchport voice vlan 5	Put the interface in voice VLAN 5
SW1# show interface e0/1 switchport	Verify switchport configuration
SW1# show interface e 0/1 status	Determine the physical status

